

SHAZ FURNITUREWALA

✉ mohammadshaz529@gmail.com | 📧 shazfurniturewala | 🎓 Scholar | 🌐 wid0gast | 🌐 shxz.in

Work Authorization: F-1 student, CPT-eligible for Summer 2027 internships.

RESEARCH FOCUS

NLP researcher with 11 peer-reviewed publications at ACL, EMNLP, AAAI, and ICWSM, several as first author, complemented by two years building and evaluating LLM systems in production at scale. Current focus: **mechanistic interpretability** of language models, adversarial robustness, fairness and human alignment, and evaluation of reasoning and agentic systems.

EDUCATION

New York University

Master of Science, Data Science.

New York, USA

May 2028

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering, Computer Science. GPA: 3.71/4.0 (Equivalent)

Rajasthan, India

July 2024

SELECTED PUBLICATIONS

- **S. Furniturewala** and A. Zubiaga. Towards Inclusive Toxic Content Moderation: Addressing Vulnerabilities to Adversarial Attacks in Toxicity Classifiers Tackling LLM-generated Content. *ICWSM*, 2027 (to appear).
- **S. Furniturewala** et al. Learning Through Dialogue: Engagement and Efficacy Matter More Than Explanations. *Findings of ACL*, 2026.
- V. Kumari, S. Gupta, **S. Furniturewala**, et al. End-to-End Knowledge Graph-Based Question Answering System for Complex Queries. *Intelligent Data Analysis*, 2026.
- **S. Furniturewala** et al. Thinking Fair and Slow: On the Efficacy of Structured Prompts for Debiasing Language Models. *EMNLP*, 2024.
- P. Banerjee, A. Java, S. Jandial, S. Shahid, **S. Furniturewala**, et al. All Should Be Equal in the Eyes of LMs: Counterfactually Aware Fair Text Generation. *AAAI*, 2024.
- G. Yeo, **S. Furniturewala**, et al. Beyond Text: Leveraging Multi-Task Learning and Cognitive Appraisal Theory for Post-Purchase Intention Analysis. *ACL Findings*, 2024.
- **S. Furniturewala** et al. Impact of Decoding Methods on Human Alignment of Conversational LLMs. *WASSA @ ACL*, 2024.
- A. Gupta, **S. Furniturewala**, et al. Rhetorical Role Labeling of Legal Documents using Transformers and Graph Neural Networks. *SemEval @ ACL*, 2023.

RESEARCH EXPERIENCE

ICWSM Global Initiative Research Fellow, Queen Mary University of London, with Prof. Arkaitz Zubiaga

2025

- Used **mechanistic interpretability** to probe fine-tuned BERT and RoBERTa toxicity classifiers under adversarial attacks on LLM-generated content, identifying attention heads that were simultaneously critical to performance and vulnerable to attack.
- Showed that **different attention heads govern vulnerability across different demographic groups**, and that suppressing these vulnerable circuits improves adversarial robustness without hurting task performance, informing more inclusive toxicity classifier design. (ICWSM 2027)

Research Intern, National University of Singapore with Prof. Kokil Jaidka

Aug 2023 – Jul 2024

- Constructed parallel human-LLM conversation corpora and developed **new stylistic and psychometric alignment metrics**; analyzed decoding strategies (beam size, nucleus sampling) and identified parameter regimes improving conversational human-likeness. (WASSA @ ACL 2024)
- Designed a structured self-critique prompting framework to **reduce demographic bias** in LLM outputs without model parameter access; demonstrated fairer generations across multiple datasets while maintaining task performance. (EMNLP 2024)
- Implemented multi-task models incorporating cognitive appraisal dimensions as auxiliary supervision to improve emotion and post-consumption behavior prediction, **outperforming text-only baselines**. (ACL 2024 Findings)
- Modeled 397 human-LLM conversations on socio-political issues; mediation analysis showed that **knowledge gain flows through users' cognitive engagement rather than explanation richness**, moderated by political efficacy. (ACL 2026 Findings)

Machine Learning Research Intern, Adobe MDSR Lab

May 2023 – Aug 2023

- Built CAFIE, a plug-and-play inference-time framework incorporating counterfactual demographic prompts to **equalize next-token distributions across groups**, achieving fairer text generation without model retraining. (AAAI 2024)

Undergraduate Research Assistant, Web Intelligence & Social Computing Lab, BITS Pilani, with Prof. Yashvardhan Sharma

2021 – 2023

- Built a **rhetorical role labeling** pipeline for Indian court judgments combining transformers with graph neural networks to model inter-sentence dependencies in legal documents. (SemEval @ ACL 2023)
- Developed an **end-to-end knowledge graph question-answering system** for complex, multi-hop queries, using T5 and Longformer with a named-entity disambiguation step to **remove the need for gold entity annotations**. (Intelligent Data Analysis 2026)

INDUSTRY EXPERIENCE

AI Research Engineer, AlphaSense

July 2024 – June 2026

- Contributed to agentic overhaul of the flagship generative search into a LangGraph RAG system (4 modes: Fast, Auto, Think Longer, Deep Research) over a **500M-doc corpus** (300K/day ingestion); built subagents for query generation, relevance eval, compliance, and report writing for generative search queries.
- Designed an **attribution verification pipeline** that grounds generated claims in retrieved evidence, cascading embedding similarity, fuzzy matching, and LLM reranking over source snippets; sustains **97% precision** over 2M+ citations/day across document, image, web, and MCP sources.
- Extended this to on-demand claim-level verification, letting users re-check any answer segment against its supporting evidence with fallback replay of the original retrieval window; **reduced user-reported hallucinations by 61%**.
- Architected a multi-stage LLM summarization pipeline using prompt-chaining and self-consistency to generate 4 structured report types, processing **30K+ earnings and 100K+ conference transcripts/quarter**, now powering 7 product teams via an internal API.

Machine Learning Intern, Clientell Technologies

May 2022 – Aug 2022

- Trained an **email topic categorization model** to classify client emails into over 30 business categories for sales and customer support.
- Built a **meeting email detection tool** that used machine learning to detect emails related to meetings and assisted in scheduling them.

SKILLS

Research: Mechanistic interpretability (attention-head attribution, circuit ablation), adversarial robustness, fairness and bias evaluation, human alignment, controlled decoding, multi-task learning, LLM-as-judge and human evaluation design.

Modeling: Encoder-only and encoder-decoder transformers, autoregressive LLMs, graph neural networks, fine-tuning with LoRA and PEFT, embeddings and representation learning.

Frameworks and Tools: PyTorch, Hugging Face Transformers, SentenceTransformers, NetworkX, Weights & Biases, Python, SQL, C++.

Systems: RAG and agentic orchestration (LangGraph, LangChain), information retrieval and reranking, vector search (Vespa), AWS (SageMaker, Bedrock, S3), Docker, Kubernetes, CI/CD.

AWARDS & HONORS

- **ICWSM Global Initiative Research Fellow**, 2025: selected as 1 of 4 applicants worldwide for a mentored fast-track research program with a senior faculty mentor.
- All India Rank 3196 / 150,838, JEE Advanced 2020 (99.98 percentile); All India Rank 77, National Science Talent Search Examination 2019.

TEACHING EXPERIENCE

- **Teaching Assistant, Graph Mining** | BITS Pilani (Aug 2023 – Dec 2023): Conducted weekly labs for 36 students on graph algorithm implementation using NetworkX; prepared and graded assignments.
- **Teaching Assistant, Neural Networks and Fuzzy Logic** | BITS Pilani (Aug 2022 – Dec 2022): Delivered introductory lecture to 117 students on forward and backward propagation; designed and graded 3 Jupyter notebook assignments; mentored 6 students on a semester project implementing a recent ML paper.

PROFESSIONAL SERVICE

- **Reviewer**, ACL Rolling Review (ARR): June 2024, February 2025, and January 2026 cycles.
- **Reviewer**, International AAAI Conference on Web and Social Media (ICWSM): 2025 (September cycle), 2026 (September and January cycles), and 2027 (May cycle).

ACTIVITIES & LEADERSHIP

BITS Pilani: Secretary, The Debating Society (2022–23); Chief Editor, Cactus Flower Magazine (2022–23); Election Commissioner, Society for Student Mess Services (2022–23).